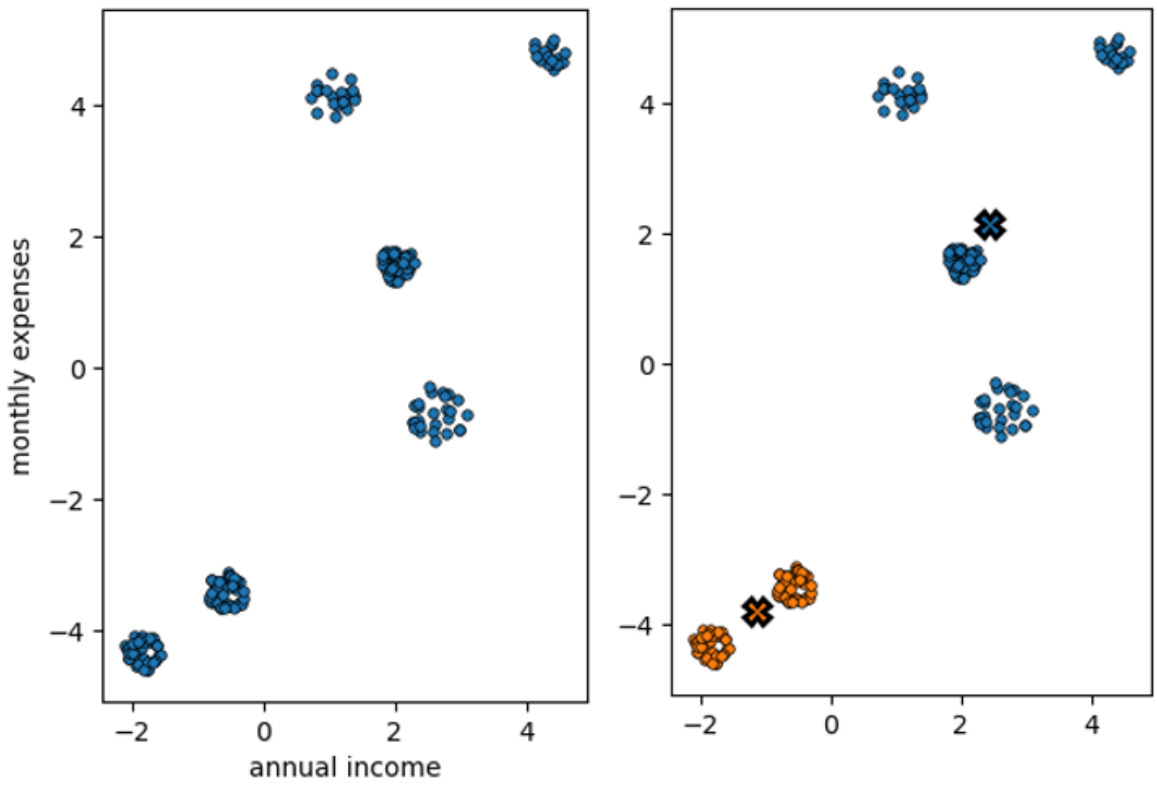


Figure 12.24 A synthetic dataset showing customers of a hypothetic e-commerce website, after feature scaling. On the right, a possible (bad) clustering for the dataset.

To avoid this effect, we perform a normalization step by subtracting the mean of each feature to every single point, and then dividing by the feature's standard deviation[25].

On the left side of figure 12.24 we show a possible clustering for the dataset: you don't need to be a domain expert to see that this appears a bad choice for the number of clusters. Can we write a method, or a mathematical formula, to express the (poor) quality of this choice?

Let's start from a consideration: what are the consequences of a bad clustering? Looking at the picture, we can see that